
Auditing VLM Benchmarks for Free: Detecting Benchmark Pathologies from Existing Evaluation Runs

Anonymous Author(s)

Affiliation

Address

email

Abstract

Every evaluation run of a vision–language model already produces per-question artifacts—predictions, judged correctness, metadata—that leaderboards reduce to a single score and then discard. We treat those artifacts as a free audit of the benchmark itself, and add an audit layer to VLMEvalKit: six detectors that run as post-processing over saved outputs, plus one cheap text-only rerun with the images stripped, emitting a per-question report in a single command. The layer reports what survives a null model rather than raw detector counts, attaching to each finding its excess over an item-matched permutation null, a confidence interval, and its stability across model pools. Auditing MMStar—a benchmark curated specifically so that the image is indispensable—with three models, we find that on a fifth of its questions the pool gains nothing from the image, and that 3.2 percentage points of that share exceed the null ($p = 0.001$), concentrated in the mathematics and science categories and absent from coarse perception. A second detector flags 134 questions on which every model that answered in parseable form picked the same non-gold option, and these peak in exactly the perception categories where the first signal disappears, so the two pathologies are separable from the same artifacts. Replacing the LLM judge with exact string matching changes no conclusion, and every scoring discrepancy between the two traces to one model whose free-form answers a parser cannot read. On a binary yes/no benchmark the detectors measurably fall out of calibration—a scope boundary we measure and report rather than leave implicit. The audit layer, the prediction artifacts and the generated reports accompany this submission.

1 Introduction

MMStar was built by filtering candidate questions so that the image is indispensable: items solvable by blind language models were removed by construction [Chen et al., 2024]. Run three VLMs on it twice—once as intended, once with every image stripped from the prompt—and align the two runs question by question. On 303 of 1,500 questions (20.2%) the pool is exactly as accurate without the image as with it. Most of that share is arithmetic rather than pathology: even blind answers unrelated to the items tie full-run accuracy sometimes, and an item-matched permutation null puts that baseline at 17.0%. The remaining +3.2 pp ($p = 0.001$) is signal, and it is not spread evenly—it sits in the two categories where world knowledge substitutes for looking at the picture. Finding this required no new benchmark, no annotation campaign, and almost no extra compute: everything except one text-only rerun was already sitting in the evaluation toolkit’s output directory.

That is the argument of this paper. A growing literature documents pathologies of multimodal benchmarks: items solvable without the image [Goyal et al., 2017, Chen et al., 2024, Huang et al.,

2024, Joshi et al., 2026], wrong gold labels [Northcutt et al., 2021, Gema et al., 2025, Vendrow et al., 2025], position and selection biases in multiple choice [Zheng et al., 2024, Liu et al., 2024, Song et al., 2024], and sensitivity of reported scores to the extraction protocol [Rosenthal et al., 2025]. The standard responses are to build cleaner replacement benchmarks [Chen et al., 2024, Li et al., 2024, Vendrow et al., 2025, Joshi et al., 2026] or standalone flaw-mining pipelines [Truong et al., 2025, Balepur et al., 2026]. Both treat auditing as a separate research project with its own inference budget. We take an infrastructure view instead: the artifacts that toolkits such as VLMEvalKit [Duan et al., 2024] and LMMs-Eval [Zhang et al., 2024] persist on every run are already sufficient to audit the benchmark as a side effect of evaluating on it.

1.1 Contributions

1. **An audit layer for an evaluation toolkit rather than a standalone pipeline.** Six benchmark-pathology detectors consume the per-question files VLMEvalKit already writes; the only added inference is a text-only rerun, implemented centrally so that local and API models are stripped of visual content identically (Section 3).
2. **A way of reading detector output that does not overstate it.** A raw flag count is not interpretable, because part of it is produced by marginal accuracies alone. We report instead the excess over an item-matched permutation null, with interval estimates, leave-one-model-out ranges and per-category strata, and show that the excess is the pool-stable quantity while the raw share is not (Section 3.3).
3. **A case study that separates two pathologies inside a curated benchmark.** On MMStar, text-solvable residue and label-error candidates are both present, are recovered from the same artifacts, and occupy opposite ends of the benchmark’s own category taxonomy (Sections 4.1–4.3).
4. **An account of the audit’s own validity and limits.** We quantify how much of the audit depends on an LLM judge, attribute the residual dependence to a single model’s output format, and identify a format—binary yes/no items—on which the detectors demonstrably lose calibration (Sections 4.4–4.5).

We do not claim to discover these pathologies; blind-solvability and label errors in VLM benchmarks are documented at scale by concurrent work [Joshi et al., 2026, Zhou et al., 2026]. Our claim is narrower: these checks can run inside the evaluation infrastructure, from artifacts it already saves, with uncertainty attached—so that every evaluation run can double as a benchmark audit.

2 Related work

Benchmark pathologies. Language priors in visual question answering were established by Goyal et al. [2017]; MMStar [Chen et al., 2024] and MMEvalPro [Huang et al., 2024] showed that blind language models score far above chance on multimodal multiple-choice suites; concurrent work probes the visual dependence of existing suites through blur ablations [Zhou et al., 2026] or phrasing-controlled items [Singla et al., 2026]. Label errors destabilize rankings [Northcutt et al., 2021]; re-annotation efforts include MMLU-Redux [Gema et al., 2025] and platinum benchmarks [Vendrow et al., 2025], and Li et al. [2025] report a fallibility analysis covering MMStar. Position and selection bias in multiple choice is established for language models [Zheng et al., 2024], handled in MMBench through CircularEval [Liu et al., 2024], and reused as a contamination probe [Song et al., 2024]. Kim et al. [2025] show that model errors correlate across vendors and scales, which caps what any cross-model consensus can establish; we return to this in Sections 4.3 and 5.

Audit pipelines and toolkits. Fantastic Bugs [Truong et al., 2025] mines response patterns to flag invalid questions in nine text benchmarks; BenchMarker [Balepur et al., 2026] applies an education-inspired rubric with LLM judges to text multiple-choice benchmarks; DatBench [Joshi et al., 2026] audits 33 VLM datasets and releases a cleaned suite. Our deliverable differs: not a cleaned dataset and not a standalone pipeline, but audit machinery embedded in a mainstream toolkit, reusable on any benchmark–model combination from artifacts the toolkit already saves. To our knowledge no such layer exists for VLMEvalKit or LMMs-Eval [Zhang et al., 2024]. Redundancy analyses over existing score matrices [Zhang et al., 2025] share the reuse-the-exhaust spirit for a different question.

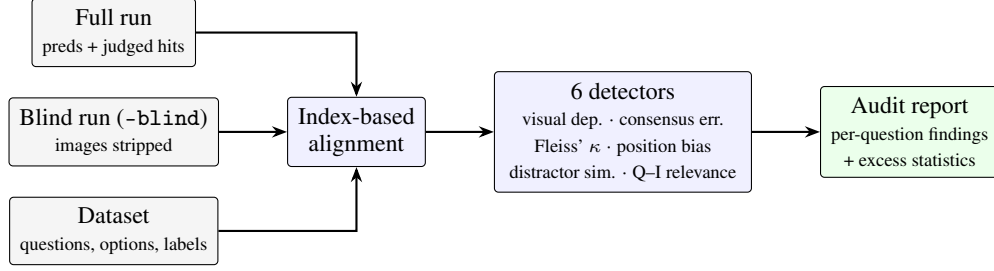


Figure 1: The audit pipeline. Standard evaluation artifacts—full runs plus one cheap text-only blind run—are aligned on dataset question ids and passed to six detectors; the output is a per-question report with permutation-null statistics and a reproducibility manifest (git commit, command-line arguments, generation parameters). Only the blind run costs inference, and it is text-only.

Table 1: Detector taxonomy. “Judge-dep.” marks detectors whose output depends on the answer-extraction protocol.

Detector	Signal	Pathology flagged	Judge-dep.
visual_dependency	full vs. blind accuracy	no-gain-from-image items	yes
consensus_error	majority vote vs. label	annotation-error candidates	no
fleiss_kappa_agreement	inter-model κ	unstable / ambiguous items	no
correctness_agreement	judged-correctness profile	difficulty / discrimination	yes
answer_options_distribution	label-position histogram	position bias	no
distractor_similarity	pairwise option similarity	near-duplicate options	no

On the measurement side, Rosenthal et al. [2025] document protocol sensitivity of VQA accuracy; we extend that question from accuracy numbers to audit conclusions.

3 Methodology

3.1 Inputs and the blind-run protocol

The layer consumes the per-question result files of n models on one benchmark, each with a raw prediction, the gold answer, and optionally a judged correctness flag. A *blind run* repeats inference with all visual content removed from every prompt, text untouched, implemented once inside the toolkit so that local and API models are treated identically; blind artifacts get a `_blind` suffix and sit beside the full run. All frames are aligned on the dataset-level question id—never on row order—and restricted to the shared id set.

Two rules protect validity. First, model answers are never back-filled from gold labels, whatever the judge decided; an earlier version of our own pipeline made exactly this mistake, and repairing it multiplied the unanimous-conflict count by eleven (Section 4.6). Second, predictions from which no option letter can be parsed become an abstention sentinel that is excluded from consensus votes rather than silently counted as wrong.

3.2 Detectors

Table 1 summarizes the six detectors. Consensus, agreement, position-distribution and distractor detectors parse option letters from raw predictions and never consult a judge; visual dependency and correctness agreement consume judged labels and are re-checked under exact matching in Section 4.4.

For visual dependency, each question is categorized from model-averaged full accuracy a_q^F and blind accuracy a_q^B : *visual-dependent* ($a_q^F \geq 0.99$ and $a_q^B \leq 0.01$), *conflicting* ($a_q^B > a_q^F + \delta$, the image hurts), *visual supplement* ($a_q^F - a_q^B \geq \delta$), and *text-only* otherwise, with $\delta = 0.2$. With $n = 3$ models the accuracies live on the grid $\{0, 1/3, 2/3, 1\}$, so text-only reduces to exact equality $a_q^F = a_q^B$ and carries no tunable threshold. The visual-dependent share, in contrast, is threshold-fragile—22.5% under the strict definition against 54.8% under a majority definition (Appendix B)—which is why

we never headline it. Consensus conflicts are tiered: *unanimous* (every parseable answer is the same non-gold option), *high* (all models disagree with the gold label), *medium* (a qualified majority disagrees).

3.3 Audit statistics: report the excess, not the share

A raw text-only share is easy to over-read. Even if blind answers were drawn independently of the items, some questions would tie full-run accuracy by chance, and that baseline moves with the pool’s marginal accuracy—across our two- and three-model pools the raw share swings by 14.2 pp. The layer therefore reports the share’s *excess over a permutation null*: blind score vectors are shuffled across questions, with the same permutation applied to every model, which preserves each model’s marginal blind accuracy and the inter-model correlation structure while destroying only the per-question full-to-blind pairing. We use 999 permutations; the one-sided p -value uses the standard $(g+1)/(N+1)$ correction, so the smallest reportable value is $p = 0.001$. The same module attaches Wilson 95% intervals to headline proportions (item-level uncertainty), leave-one-model-out pools (pool sensitivity), a split-half reproducibility check, and a stratification by official dataset categories (localization and internal replication). All of it is post-processing; no model is queried again.

3.4 Evaluation protocol

Benchmarks. MMStar [Chen et al., 2024]: 1,500 four-option multiple-choice items curated for vision-indispensability, with six official categories of 250 items each. HallusionBench [Guan et al., 2024]: 951 yes/no items built around visual illusions and language priors, used here as a deliberate out-of-format probe of the detectors themselves.

Models. InternVL3-8B, Ristretto-3B and gpt-5-nano ($n = 3$): two open models and one API model. This is a case study sized to validate the audit machinery, not a leaderboard; leave-one-model-out ranges are reported throughout and are wide (Table 2).

Scoring. The judged arm uses gpt-4o-mini as the answer-extraction judge, identical for full and blind runs; the exact arm parses option letters directly from predictions with no model in the loop. Consensus, κ , position and distractor detectors use the exact arm by construction. Both arms are reported wherever they can differ.

Reproducibility. Every run records its git commit, command-line arguments and generation parameters in a manifest; the audit is deterministic given the artifacts and a fixed permutation seed. The audit layer, the prediction artifacts for both benchmarks, and the generated reports accompany this submission; Appendix A gives the exact commands that regenerate every number and figure in this paper.

4 Results

4.1 A curated benchmark still carries a text-solvable excess

Figure 2 and Table 2 give the headline result. The observed text-only share is 20.2% (Wilson 95% CI [18.2, 22.3]); the permutation null predicts 17.0%; the excess is +3.2 pp at $p = 0.001$, the floor of a 999-permutation test. The excess survives every leave-one-model-out pool (+3.6 to +7.7 pp, each $p = 0.001$) and its split-half reproducibility is ± 1.5 pp. Normalizing by the null is what makes the number stable: the raw share varies by 14.2 pp across pools, the excess by 4.6 pp.

Two readings of this result must be kept apart, and the per-item composition decides between them (Figure 2c). Of the 303 no-gain questions, 164 are items on which every model is wrong both with and without the image—evidence of shared failure, not of text-solvability. Only 26 are answered correctly by all three models blind. The claim we make is therefore about zero marginal value of the image for this pool, and the excess quantifies how much of that is more than marginal-rate arithmetic predicts. We also read the result as evidence that MMStar’s curation largely worked: unfiltered suites show blind accuracy tens of points above chance [Chen et al., 2024, Huang et al., 2024], whereas the residue here is 3.2 points above the matched null. The audit’s role is to measure what survives curation, cheaply and per item.

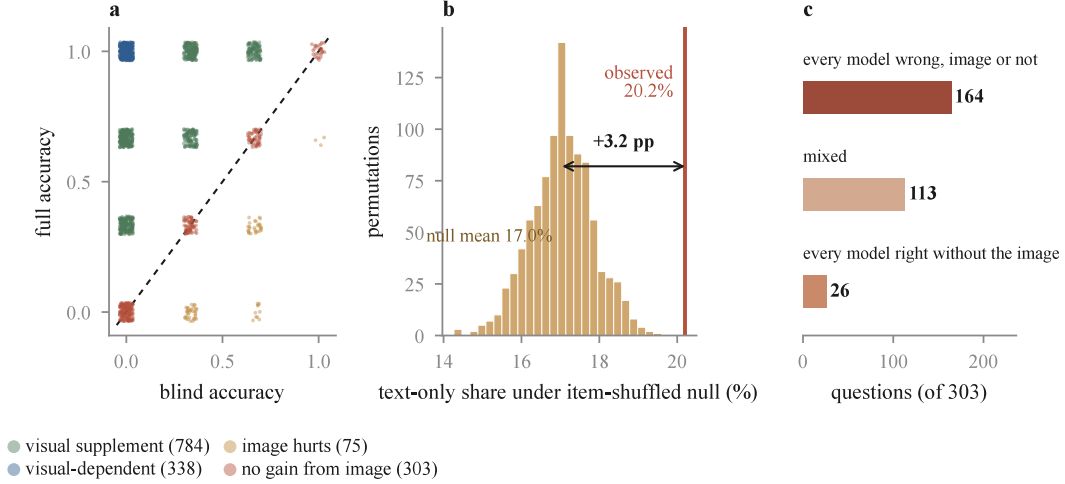


Figure 2: The text-only signal on MMStar. **(a)** Model-averaged full against blind accuracy per question (judged arm, jittered; three models give the 4×4 grid). Questions on the diagonal gain nothing from the image. **(b)** The observed 20.2% share against 999 item-shuffled permutations: the null centres at 17.0%, leaving a +3.2 pp excess that no permutation reaches ($p = 0.001$). **(c)** The no-gain set is not uniform—on 164 of 303 questions every model is wrong with and without the image, so the honest reading is *zero marginal value of the image*, of which genuinely text-answerable items are one part.

Table 2: Text-only share against the permutation null on MMStar (999 permutations, one-sided, $(g+1)/(N+1)$ correction). The excess, not the raw share, is the pool-stable quantity: raw spread 14.2 pp, excess spread 4.6 pp.

Model pool	Observed (%)	Null mean (%)	Excess (pp)	p
All three models	20.2	17.0	+3.2	0.001
without InternVL3-8B	27.9	24.1	+3.7	0.001
without Ristretto-3B	25.5	21.9	+3.6	0.001
without gpt-5-nano	34.4	26.7	+7.7	0.001

4.2 The two pathologies live in different places

Stratifying by MMStar’s six official categories turns one number into a pattern (Figure 3). The text-only excess concentrates where world knowledge can substitute for perception: math (+6.6 pp) and science & technology (+7.5 pp). Coarse perception shows none (−1.5 pp)—blind models cannot fake seeing. Unanimous consensus conflicts distribute the other way around: they peak in the perception categories (11.6% coarse, 14.4% fine-grained) and are lowest in math and reasoning (4.4–6.8%), with science & technology intermediate (10.4%). The two detectors are not renamings of one defect; they flag largely disjoint subsets with opposite category profiles.

4.3 Label-error candidates

The consensus detector flags 23.9% of MMStar items at some tier; the tier that matters is the unanimous one. On 134 questions (8.9%, Wilson 95% CI [7.6, 10.5]) every parseable model answer is the same option and it is not the gold label; on 117 of those all three models answered in parseable form, and the remaining 17 rest on two, because the abstention rule excludes rather than guesses. Fleiss’ κ over raw extracted answers is 0.495 (0.544 when items with abstentions are excluded), so unanimity is not the baseline behaviour of this pool. The rate is of the same order as the annotation-error rates reported for MMStar by the independent fallibility analysis of Li et al. [2025], which did collect fresh human labels.

We deliberately call these *candidates*, and the reason is measurable in our own pool. Errors of different models are correlated [Kim et al., 2025]: when two of our models are both wrong, they

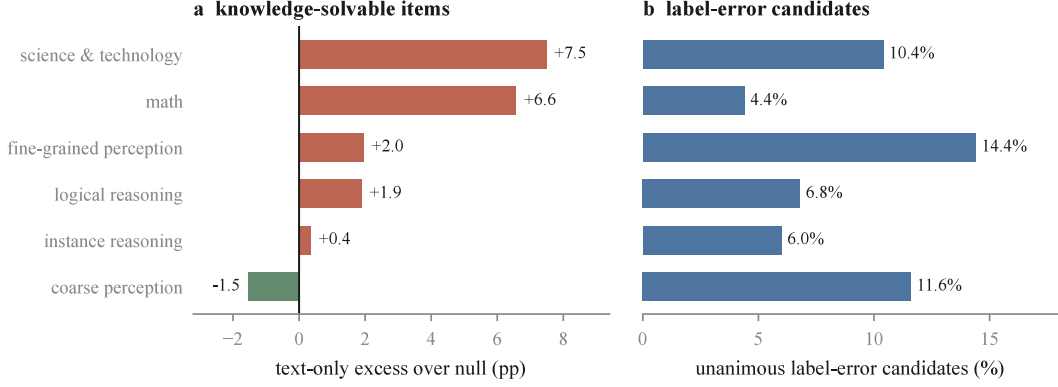


Figure 3: Per-category audit of MMStar (250 items per category), sorted by text-only excess. **(a)** The excess concentrates where world knowledge can substitute for perception and vanishes in coarse perception. **(b)** Unanimous label-error candidates order themselves almost in reverse, peaking in the perception categories. The two detectors flag different parts of the benchmark; six categories moving coherently also act as internal replication.

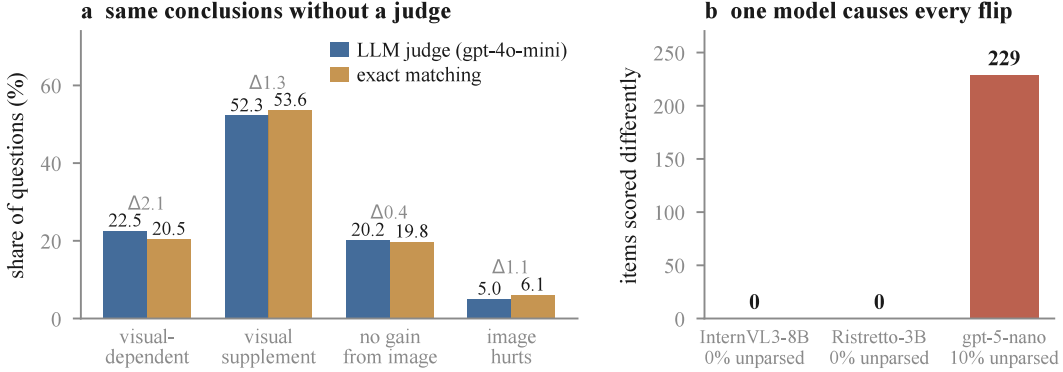


Figure 4: Judge dependence of the audit on MMStar. **(a)** Category shares under the gpt-4o-mini judge and under exact letter matching, on identical predictions: every share moves by at most 2.1 pp and no conclusion changes. **(b)** The disagreement has a single source—for the two open models the judge and the letter parser score every item identically, while gpt-5-nano, whose answers are unparsable on 10% of items, accounts for all of it.

179 pick the same wrong option in 64–74% of cases, against 26–27% when the item-to-item pairing is
 180 destroyed by the same shuffle that defines our null. Unanimity therefore cannot adjudicate a gold
 181 label on its own—it concentrates human review, which remains the arbiter. The audit layer ships the
 182 flagged list with per-question model votes for exactly that purpose; a human verification campaign
 183 over a stratified sample is the natural next step and is not part of this paper’s claims.

184 4.4 Removing the LLM judge changes almost nothing, for a reason

185 Audit claims should not silently inherit an LLM judge [Rosenthal et al., 2025]. Four of six detectors
 186 never use one (Table 1). For the two that do, we recomputed everything under exact letter matching
 187 on identical predictions (Figure 4a): category shares move by at most 2.1 pp (text-only 19.8% against
 188 20.2%; visual-dependent 20.5% against 22.5%; supplement 53.6% against 52.3%; conflicting 6.1%
 189 against 5.0%), and nothing in Sections 4.1–4.3 changes.

190 The instructive part is where the two arms disagree. 122 of 1,500 items (8.1%) change visual-
 191 dependency category between arms, and every one of them traces to gpt-5-nano: the judge and the
 192 letter parser score that model differently on 229 model-item pairs, and the two open models on none
 193 at all (Figure 4b). The mechanism is prosaic—gpt-5-nano answers in free-form prose on 10% of
 194 items, which a regular expression cannot read but a judge can. The judge is thus a robustness arm

rather than a dependency: for models with canonical outputs exact matching suffices, and for models without, the audit report can say so per model instead of averaging the discrepancy away.

4.5 Where the detectors stop working: a binary-format probe

We ran the same audit on HallusionBench [Guan et al., 2024], whose 951 items are yes/no questions, expecting trouble—and measuring it. With two answer options blind guessing sits at 50%, unanimity is cheap, and the multiple-choice consensus machinery does not apply at all: option letters do not parse and the exact arm reports abstention everywhere. The text-only excess is still large and significant (+5.4 pp over an 18.5% null, $p = 0.001$), but the decisive calibration test fails. Ranking items by their continuous visual gain recovers HallusionBench’s own visual-dependent against visual-supplement design labels at AUC 0.49, that is, at chance (Appendix C). One directional signal survives and matches the benchmark’s intent: the excess is 2.3 times larger on design-VS items (+7.7 pp) than on design-VD items (+3.3 pp). We conclude that the current detector suite is calibrated for k -way multiple choice and treat binary formats as out of scope—a measured boundary rather than a footnote, since an audit tool that does not state where it is invalid invites the same over-reading it exists to prevent.

4.6 Auditing the audit

Audit pipelines can manufacture their own findings, so we checked ours against the known failure modes. *No-op perturbation*: the blind run strips images centrally, blind transcripts contain model statements that no image is present, and blind accuracy falls towards the option prior in the perception categories. *Extraction artifacts*: the judged-against-exact arm (Section 4.4) measures precisely this and attributes all discrepancy to one model. *Unfaithful scoring*: the judged arm is VLMEvalKit’s native protocol, unmodified. *Broken pairing in uncertainty*: the permutation null is item-paired by construction—only the full-to-blind linkage is destroyed. *One scalar, two readings*: every headline is disaggregated (category strata, per-pool tables, and the 164-of-303 composition in Figure 2c). One bug we did catch and fix is reported in Section 3.1: back-filling gold labels into model answers had suppressed unanimous conflicts from 134 to 12, and the layer now refuses to do it.

5 Limitations

(i) Three models. Consensus unanimity among correlated models [Kim et al., 2025] is weak per-item evidence—we measure 64–74% same-wrong-option agreement—so the detector produces candidates for review rather than verdicts, and the leave-one-model-out ranges in Table 2 show how much three models leave open. (ii) The label-error candidates in this paper are unverified by humans; the precision of the consensus detector is therefore unmeasured, and the 8.9% figure is a candidate rate, not an error rate. (iii) Unparseable predictions become abstentions, which bakes each model’s refusal policy into per-item statistics. (iv) A blind-run gain conflates language priors, answer-option statistics and training-data contamination; the audit bounds their sum and cannot separate them (cf. Song et al., 2024). (v) The visual-dependent and supplement categories depend on the threshold δ ; we headline only threshold-free quantities and publish the sweep (Appendix B). (vi) Binary yes/no formats are out of calibration, as measured in Section 4.5. (vii) One benchmark is audited in depth; the claims are about the audit machinery, and its base rates should not be extrapolated to other suites without running it.

6 Conclusion

An evaluation toolkit already stores what a benchmark audit needs. Adding one text-only rerun and a statistics pass turns those artifacts into per-item findings with uncertainty attached: on MMStar, a +3.2 pp text-solvable excess concentrated in knowledge categories, and 134 unanimous label conflicts concentrated in perception categories. The practical implication runs through leaderboards: a model delta smaller than a benchmark’s measured artifact rates should not be read as a capability difference, and an audit report costs little enough to ship alongside every score. We are extending the pool to six models, running the human verification the consensus detector calls for, and sweeping the remaining VLMEvalKit benchmarks; the layer, artifacts and reports are released so that others can do the same.

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 316 *arXiv preprint arXiv:2606.06890*, 2026. URL <https://arxiv.org/abs/2606.06890>.

317 A Reproducing every number in this paper

318 The audit consumes only the per-model result files that VLMEvalKit writes for a normal evaluation,
 319 plus the matching blind-run files. With those in place, two commands regenerate all reported statistics
 320 and all figures:

```
321 python scripts/audit_significance.py \
322     -full <model>_MMStar_gpt-4o-mini_result.xlsx (one per model) \
323     -blind <model>_MMStar_gpt-4o-mini_result_blind.xlsx \
324     -models InternVL3-8B Ristretto-3B gpt-5-nano \
325     -strata-column category -perms 999 -seed 7 -out <report dir>

326 python scripts/audit_paper_figures.py \
327     -full ... -blind ... -models ... -out paper/figures
```

328 The blind artifacts themselves come from a normal evaluation invocation with the `-blind` flag, which
 329 is the only step that requires inference:

```
330 python run.py -data MMStar -model <model> -blind
```

331 The figure script imports its detector logic directly from the statistics script, so a figure cannot
 332 disagree with a reported number. Both scripts depend only on pandas/openpyxl (plus matplotlib
 333 for figures) and do not import the toolkit itself, so an audit can be re-run on a machine that cannot
 334 host the models. Randomness enters only through the permutation seed (default 7), which is recorded
 335 in the report.

336 **Answer extraction (exact arm).** A prediction is mapped to an option letter by three ordered rules:
 337 a bare letter, possibly punctuated; a letter followed by a delimiter at the start of the string; or a letter
 338 following the words “answer” or “option”. Anything else becomes the abstention sentinel Z, which
 339 is excluded from consensus votes. This mirrors the toolkit’s own extractor, so the audit does not
 340 introduce a second, divergent notion of what a model answered.

341 **Permutation null.** For each of 999 permutations, one random permutation of question ids is drawn
 342 and applied to the blind hit vectors of *all* models simultaneously. This preserves each model’s
 343 marginal blind accuracy and the correlation between models’ blind answers, and destroys only
 344 the per-question pairing between full and blind runs. The text-only share is recomputed on each

345 permuted dataset with the identical category rule; the reported excess is observed minus null mean,
 346 and the one-sided p -value is $(g+1)/(N+1)$, where g counts permutations with a share at least as
 347 large as observed. Stratified nulls repeat the procedure within each category. For the split-half
 348 check, questions are randomly halved, the excess is computed in each half, and the median absolute
 349 difference across resplits is reported: ± 1.5 pp on MMStar and ± 2.1 pp on HallusionBench.

350 B Threshold sensitivity of the visual-dependent share

351 The visual-dependent share depends strongly on its definition, which is why the paper never headlines
 352 it. On MMStar (judged arm), the strict all-or-none definition flags 338 items (22.5%), while a majority
 353 definition—at least two of three models correct with the image and at most one without—flags 822
 354 items (54.8%). On HallusionBench the same pair is 140 (14.7%) against 478 (50.3%). The text-only
 355 category has no threshold at $n = 3$, since the accuracy grid makes it exact equality, and the unanimous
 356 consensus count has none at any n ; those are the quantities the paper reports.

357 C Binary-format calibration check

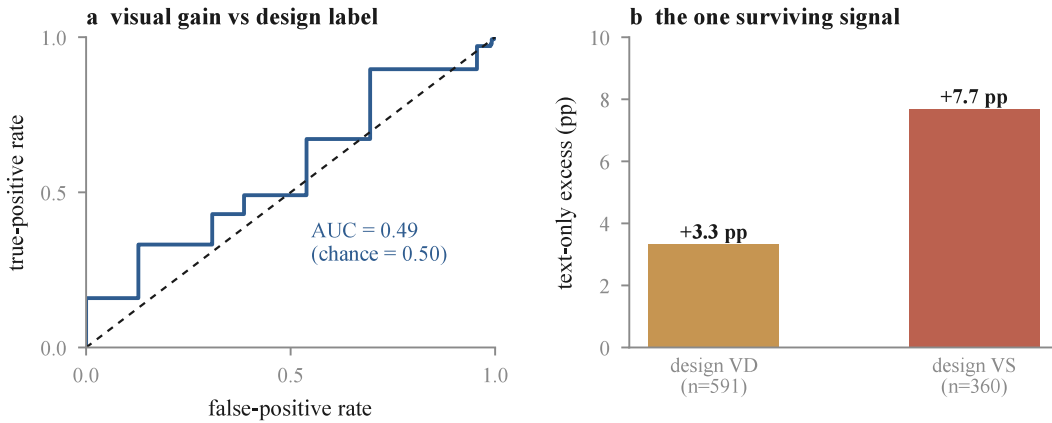


Figure 5: HallusionBench (951 yes/no items, same three models). **(a)** Ranking items by the continuous visual gain does not recover the benchmark’s own design labels (AUC 0.49 against 0.50 for chance). **(b)** The one signal that does survive points the right way: the text-only excess over the permutation null is 2.3 times larger on items the benchmark designates as visual-supplement than on those it designates as visual-dependent.

358 Design labels are read from the question index prefix that HallusionBench itself assigns (VD for visual-
 359 dependent items, the remainder treated as visual-supplement), giving 591 positives and 360 negatives.
 360 The AUC is computed on the continuous per-question visual gain $a_q^F - a_q^B$, with ties counted as half
 361 wins. Together with the degenerate letter-extraction behaviour described in Section 4.5, this is the
 362 evidence for excluding binary formats from the detectors’ current calibration.